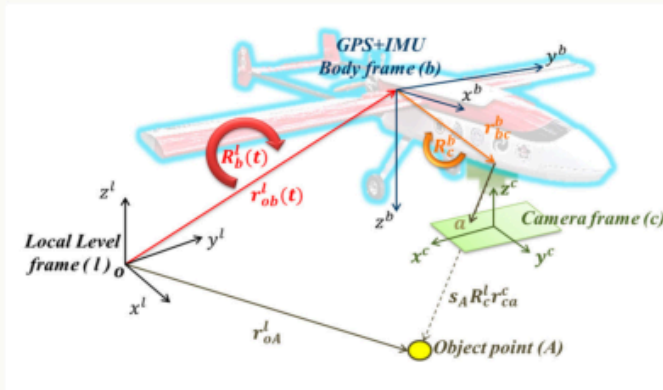
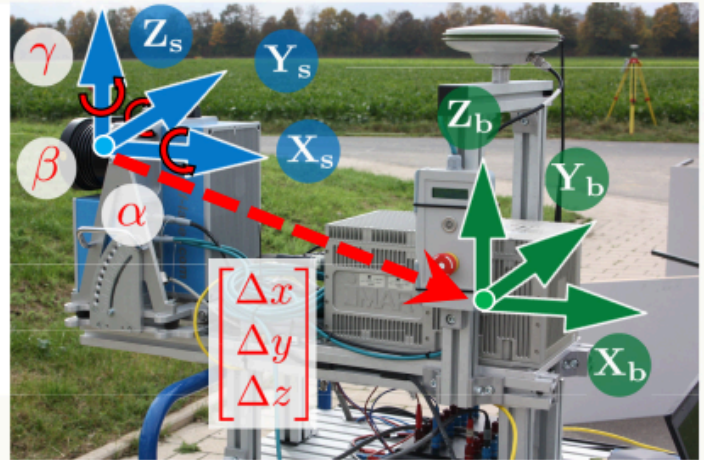


L5: System Calibration

$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \cdot \left[\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha, \beta, \gamma) \cdot \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix} \right]$$



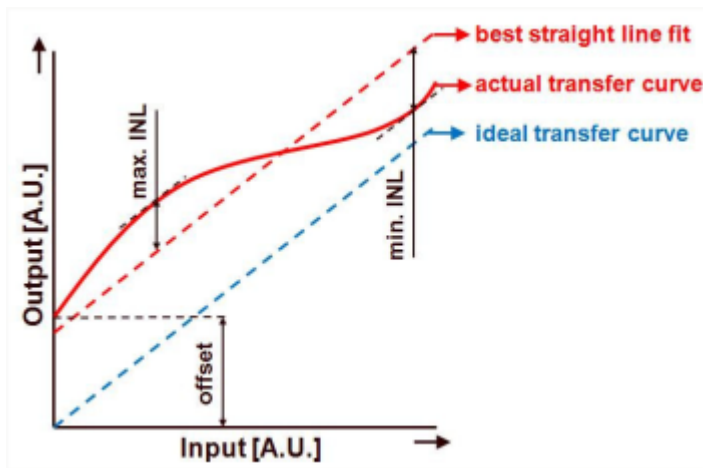
From: Chiang, K.-W.; Tsai, M.-L.; Naser, E.-S.; Habib, A.; Chu, C.-H. New Calibration Method Using Low Cost MEM IMUs to Verify the Performance of UAV Borne MMIS Payloads. Sensors 2015, 15, 5550–5595.



Formal Definition

Operation that, under specified conditions, in a first step, establishes a **relation between the quantity values** with measurement uncertainties provided by measurement standards **and corresponding indications** with associated measurement uncertainties (of the calibrated instrument or secondary standard) and, in a second step, uses this information to establish a relation **for obtaining a measurement result from an indication**

- Derive necessary parameters to receive useful information from a measurement system
- Procedure to minimise systematic errors in a measurement system
- Calibration parameters:
 - Constants
 - Parameters of functional relationships
 - Look-up tables



Examples

INERTIAL SENSORS

$$a_{P,k} = \mathbf{MS}(a_{S,k} - b_a)$$

True value

Sensor readings

Offset/Bias

$$\mathbf{M} = \begin{pmatrix} 1 & -\alpha_{yz} & \alpha_{zy} \\ 0 & 1 & -\alpha_{zx} \\ 0 & 0 & 1 \end{pmatrix}$$

Axis Misalignment

$$\mathbf{S} = \begin{pmatrix} s_{xx} & 0 & 0 \\ 0 & s_{yy} & 0 \\ 0 & 0 & s_{zz} \end{pmatrix}$$

Scale factors

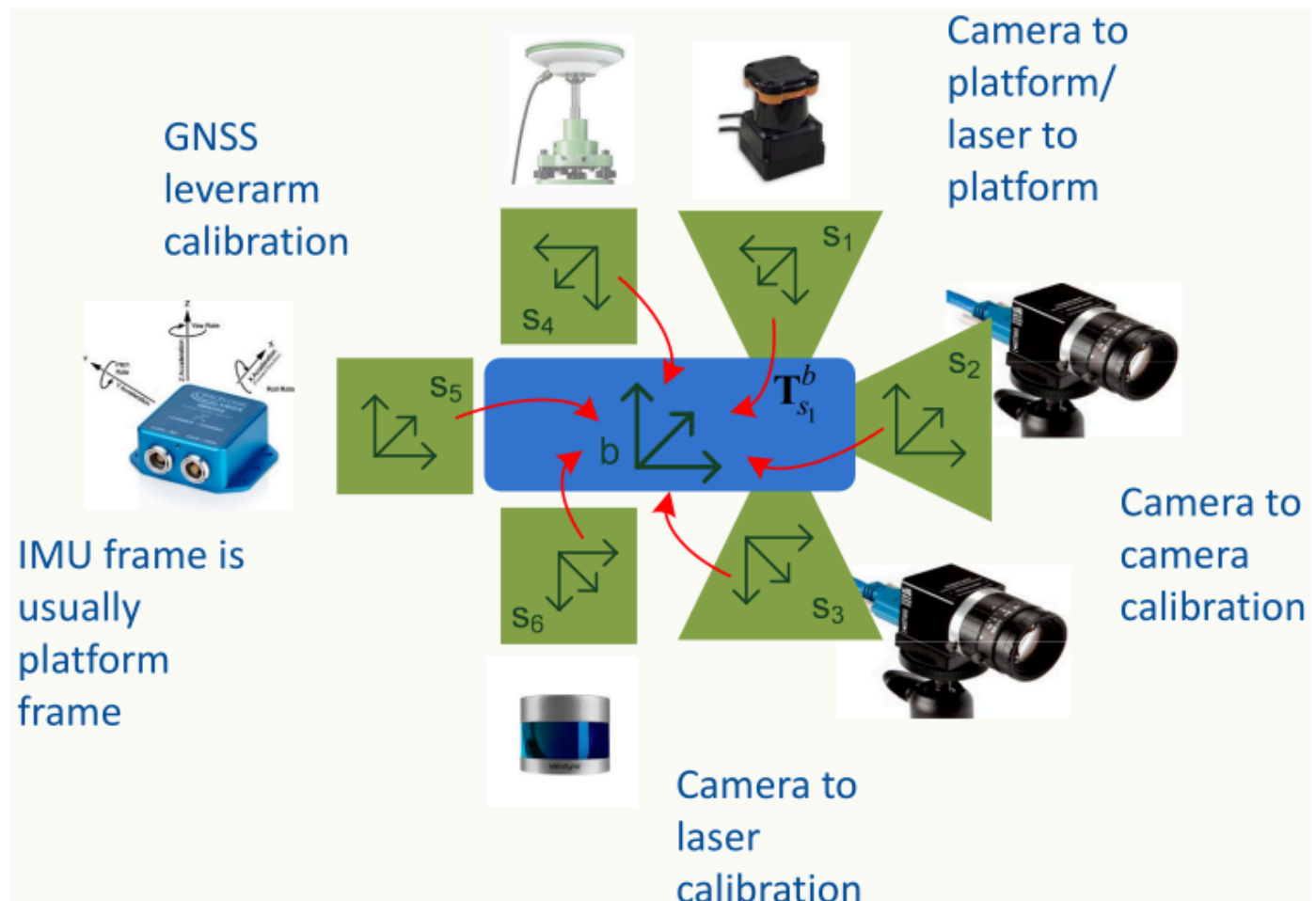
- Parameters also "calibrated over temperature"
- GNSS Antennas:
 - Additive corrections to phase observations
 - depending on:
 - Azimuth & Zenith angles of the satellites
- Laser Scanner
 - Determination of misalignments between various internal components
 - Distance measurement offsets of laser
 - ...
- Camera Calibration
 - Intrinsic camera parameters K
 - Principal point
 - Camera constant

- Scale
- Sheer

Calibration General

- Individual Calibration of single sensors or sensor groups
 - **Component calibration**
- Calibration of geometric relationships between sensors
 - **System Calibration**
 - Necessary in complex multi-sensor systems
- Temporal relationship between sensor readings
 - **Sensor synchronisation**

System Calibration



- Methods
 - Data from construction plan
 - Possibly wrong: especially parameters such as angles
 - Direct observation

- e.g. Parameters from accurate laser scan
 - Problems: Many points not interesting (e.g. camera focal point)
- Geodetic measurement equipment to derive parameters
 - Accurate but extensive
 - Problems with non physical points of interest and orientation (cameras)
- Indirect Measurements
 - **Sensors**
 - Position/orientation parameters needed
 - **Observe Objects**
 - Points, Planes, Spheres
 - **with known positions/orientations**
 - In or relative to **body frame**
- Entropy Based Methods
 - Pointcloud as Gaussian Mixture Model
 - Quadratic Renyi Entropy (QRE): Measure for quality/compactness/crispness
 - Generate PointCloud: Same areas from multiple viewing conditions
 - Assume: world not too noisy
 - Minimise QRE by varying calibration parameters
- Indirect Measurements: Plane Based
 - Arbitrary or dedicated plane setup
 - Adjustment procedure to derive parameters

Plane Based Calibration

- Laser to GNSS/IMU (=Laser to body)



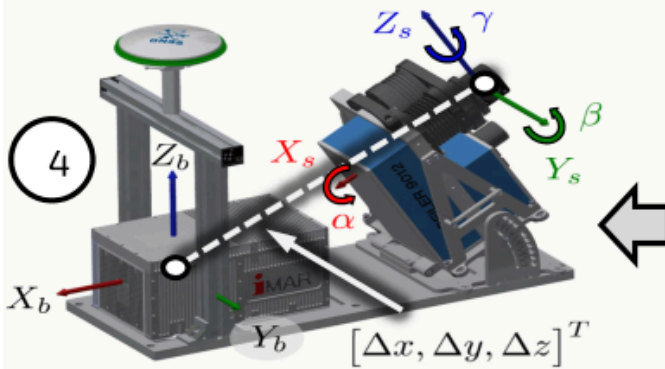
Setup of planes



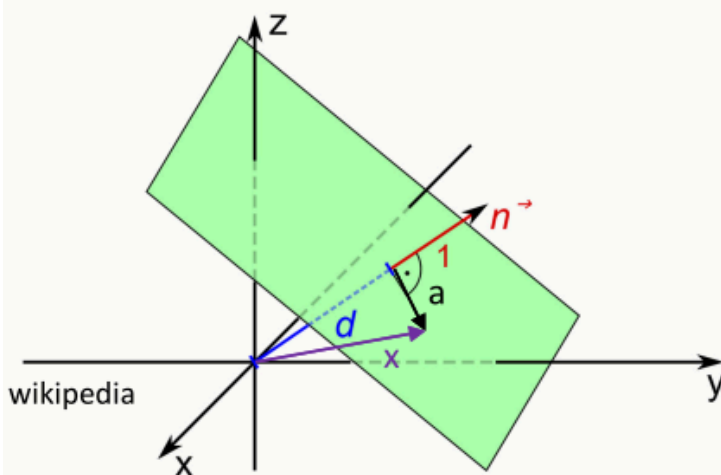
Reference scan of planes



Scan with mobile scanning system



Adjustment to derive parameters



$$\mathbf{a} \cdot \mathbf{n} = 0$$

scalar product

$$\mathbf{x} \cdot \mathbf{n} = d$$

Hesse normal form

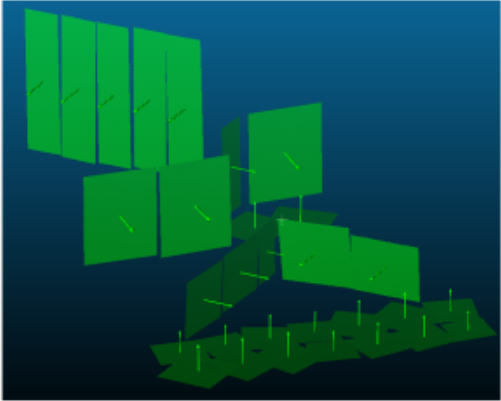
A plane can be defined by 4 parameters n_x, n_y, n_z, d

1. Estimate Plane Parameters from TLS Scans

- Implicit functional relationship between parameters and observations
- Adjustment via Gauss-Helmert model /Least Squares
- Get n, d

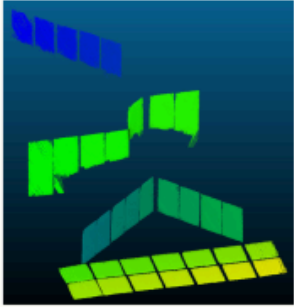
$$\begin{bmatrix} x_{TLS} \\ y_{TLS} \\ z_{TLS} \end{bmatrix}^T \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} - d_p \stackrel{!}{=} 0$$

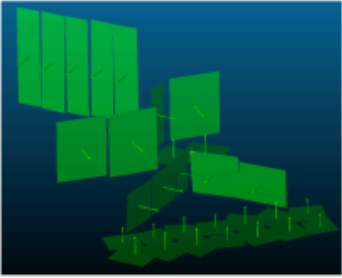
observations
parameters




2. Measure Planes with Mobile System

$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \cdot \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha, \beta, \gamma) \cdot \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix}$$



$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix}^T \underbrace{\begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}}_{\text{parameters from TLS plane adjustment}} - d_p \stackrel{!}{=} 0$$



3. Plane Based Calibration

- Implicit functional relationship between parameters and observations
- Adjustment via Gauss-Helmert model/Total Least Squares to get **Calibration Parameters**

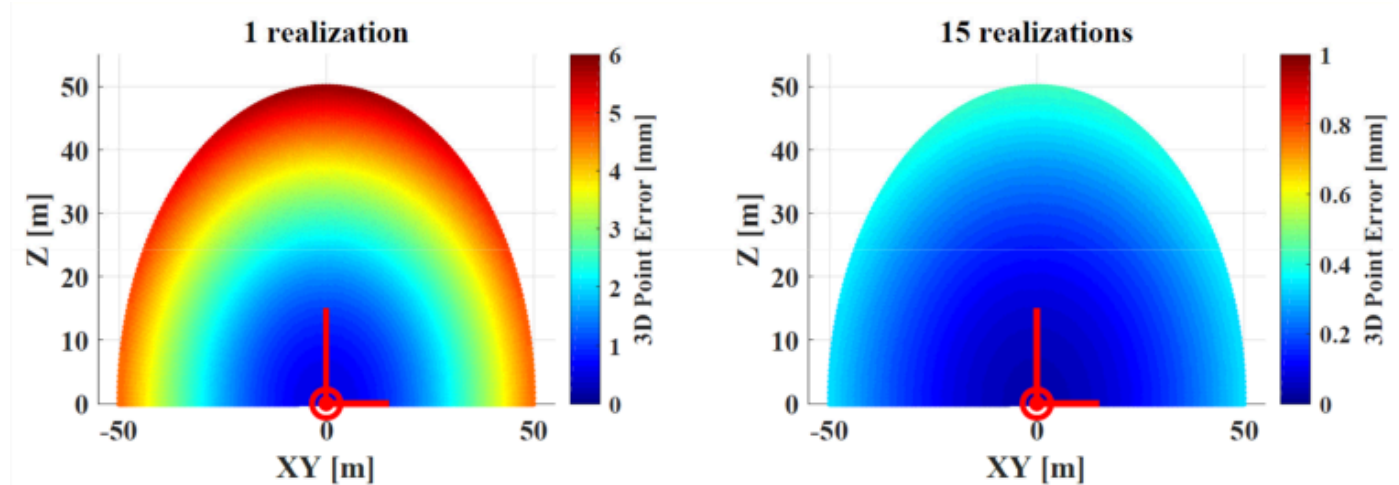
$$\left[\begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \cdot \left(\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha, \beta, \gamma) \cdot \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix} \right) \right]^T \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} - d_p \stackrel{!}{=} 0$$

observations
parameters

- Plane Parameters could also be part of the estimated parameter vector
- Dedicated plane setup is not necessary
- **Disadvantage:**
 - General Success and accuracy not guaranteed
 - Some Parameters may not be observable

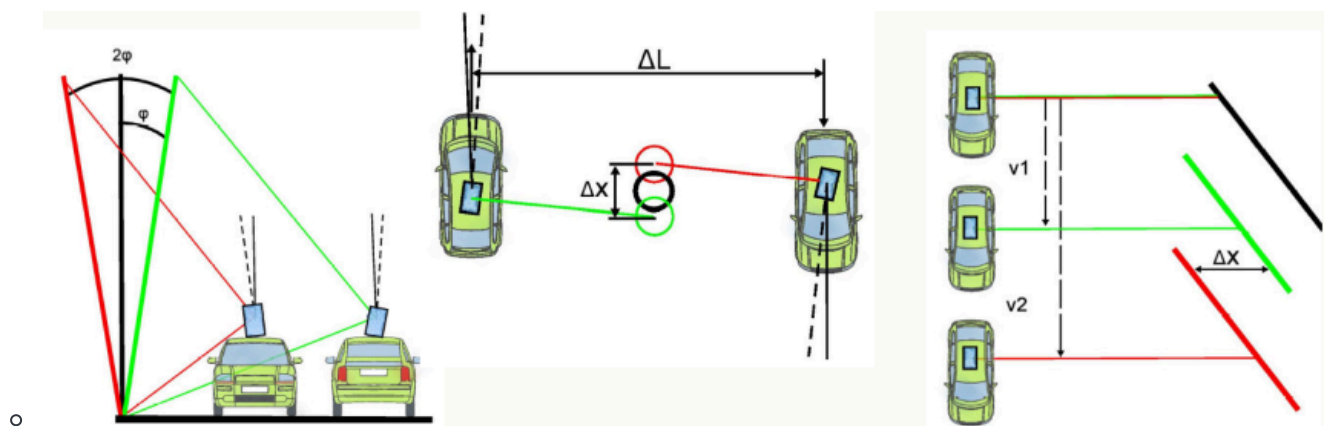
Impact of Calibration Errors

- 3D Point error from simulated calibration errors

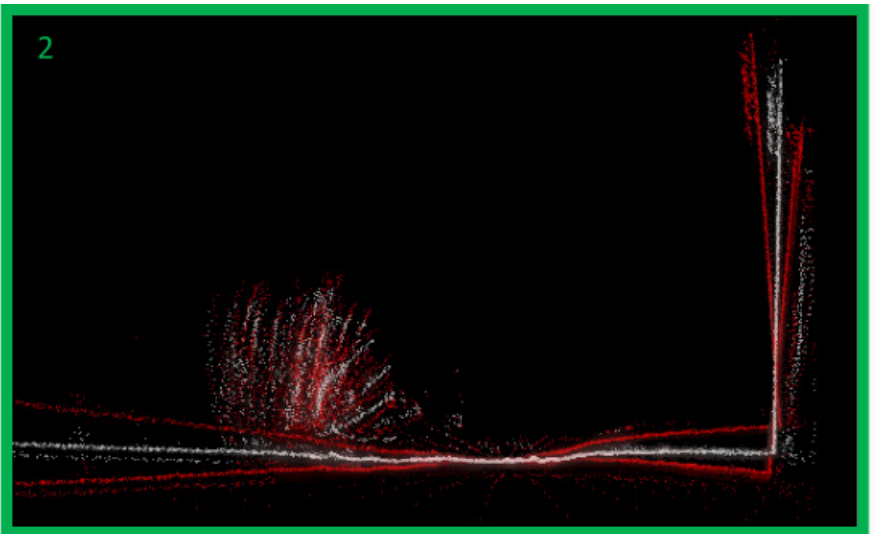
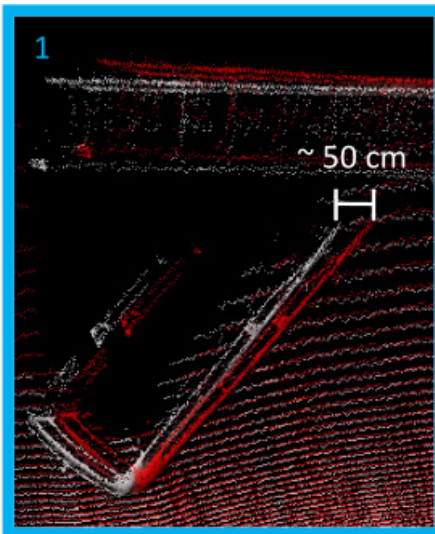
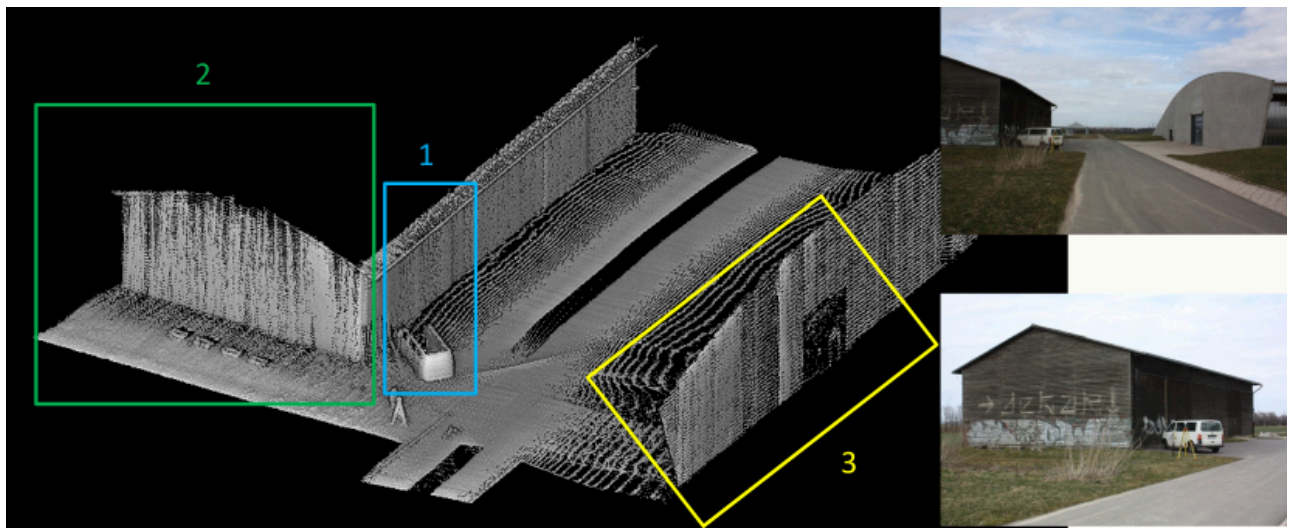


Parameter	$\sigma_{\Delta x}$	$\sigma_{\Delta y}$	$\sigma_{\Delta z}$	σ_{α}	σ_{β}	σ_{γ}
Target Accuracy	1.0 mm	1.0 mm	1.5 mm	0.0050°	0.0050°	0.0050°
1 Realization	0.9 mm	1.1 mm	4.5 mm	0.0012°	0.0188°	0.0166°
15 Realizations	0.2 mm	0.3 mm	1.2 mm	0.0003°	0.0049°	0.0043°
98 Realizations	0.1 mm	0.1 mm	0.5 mm	0.0001°	0.0019°	0.0017°

- Special driving maneuvers past building facades or artificial reference structure (e.g. planes or cylinders)

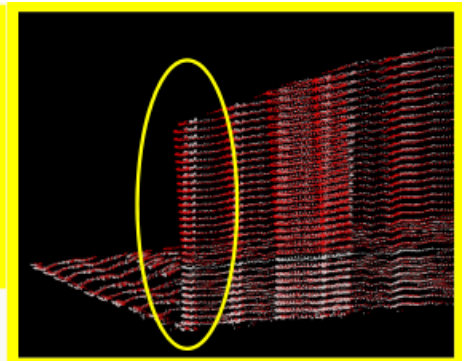
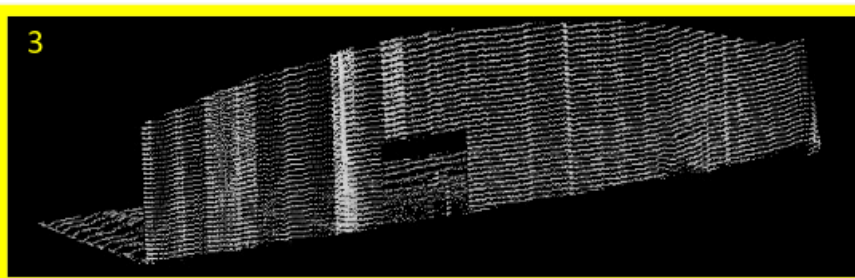


- Impact of errors in the boresight angles: $\Delta\alpha$, $\Delta\beta$, $\Delta\gamma$

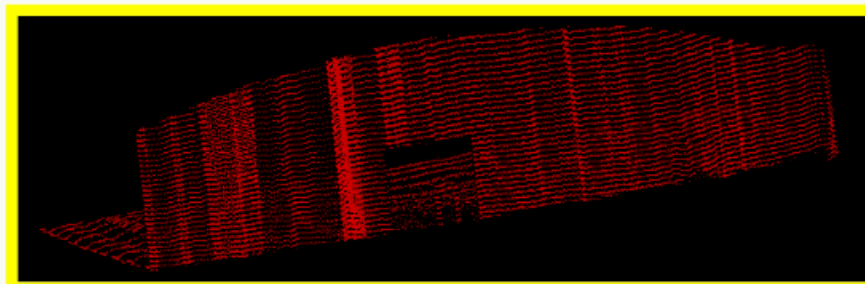


$\Delta\gamma = 5^\circ$ $s = 5,7\text{ m}$
 $\rightarrow \Delta p = 50\text{ cm}$

$\Delta\beta = 5^\circ \rightarrow$ Tilting of the facade when scanning in two different directions)



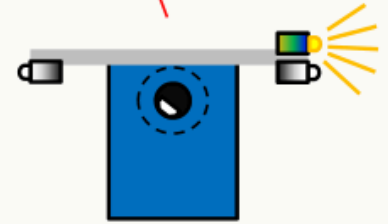
$\Delta\alpha = 5^\circ \rightarrow$ Shearing of facade



Camera to Laser Calibration

CAMERA TO LASER CALIBRATION

cameras



front view

- Checker board from camera calibration
- Find correspondences between laser and profiles and images
- Compute translation and rotation between both systems by using known point relationships